



PY32C642 Datasheet

32 bits ARM® Cortex® - M0+



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Features

- Core
 - 32 bits ARM® Cortex® - M0+
 - Up to 24 MHz as a maximum frequency
- Memories
 - 24 Kbytes Flash memory
 - 3 Kbytes SRAM
- Clock system
 - Internal 24 MHz RC Oscillator (HSI)
 - Internal 32.768 KHz RC Oscillator (LSI)
 - 32.768 KHz low speed crystal oscillator (LSE)
 - External clock input
- Power management
 - Operating voltage: 1.7 ~ 5.5 V
 - Low power modes: Sleep/Stop
 - Power-on/Power-down reset (POR/PDR)
 - Brown-out reset (BOR)
- General purpose input and output (I/O)
 - Up to 18 I/Os , all available as external interrupts
- 1 x 12 bits ADC
 - Support up to 8 external input channels and 2 internal channels
 - Four reference sources:
 - Built-in 1.5 V/2.048 V/2.5 V reference voltage,
 - supply voltage V_{CC}
- Timer
 - A 16 bits advanced control timer (TIM1)
 - A general purpose 16-bit timer (TIM14)
 - A low-power timer (LPTIM), supports wake-up from stop mode
 - An Independent Watching Timer (IWDG)
 - A SysTick timer
- Hardware CRC-32 module
- Two comparators
- Unique UID
- Serial wire debug (SWD)
- Working temp.: -40 ~ 85 °C
- Package: QFN20, QFN16

Content

1. Introduction.....	5
2. Functional overview	7
2.1. Arm® Cortex®-M0+ core	7
2.2. Memory	7
2.3. Boot mode.....	7
2.4. Clock System	8
2.5. Power Management.....	9
2.5.1. Power block diagram.....	9
2.5.2. Power monitoring.....	9
2.5.3. Voltage regulator	10
2.5.4. Low power mode	10
2.6. Rest.....	10
2.6.1. Power reset	10
2.6.2. System reset.....	11
2.7. General purpose input and output (GPIO)	11
2.8. Interrupt.....	11
2.8.1. Interrupt controller NVIC.....	11
2.8.2. External interrupt/event controller (EXTI).....	12
2.9. Analog to digital converter (ADC)	12
2.10. Comparator (COMP)	12
2.10.1. COMP main features.....	13
2.11. Timer	13
2.11.1. Advanced Timer	13
2.11.2. General purpose timer.....	14
2.11.3. Lower power timer	14
2.11.4. IWDG.....	14
2.11.5. SysTick timer	15
2.12. SWD	15
3. Pin configuration	16
3.1. Port A multiplexing function mapping	19
3.2. Port B multiplexing function mapping	19
3.3. Port C multiplexing function mapping	19
4. Memory map.....	20
5. Electrical characteristics	23
5.1. Test conditions	23
5.1.1. Min and Max.....	23
5.1.2. Typical value.....	23
5.2. Absolute maximum ratings.....	23
5.3. Operating conditions	24

5.3.1.	General operating conditions	24
5.3.2.	Power on and down operating conditions	24
5.3.3.	Embedded reset module characteristics	24
5.3.4.	Operating current characteristics	25
5.3.5.	Wakeup time from low-power modes	26
5.3.6.	External clock source characteristics	26
5.3.7.	Internal high frequency clock source HSI characteristics	28
5.3.8.	Internal low frequency clock source LSI characteristics	29
5.3.9.	Memory characteristics	29
5.3.10.	EFT characteristics	30
5.3.11.	ESD & LU characteristics	30
5.3.12.	Port characteristics	30
5.3.13.	NRST pin characteristics	31
5.3.14.	ADC characteristics	31
5.3.15.	Comparator characteristics	32
5.3.16.	Temperature sensor characteristics	33
5.3.17.	Internal reference voltage characteristics	33
5.3.18.	ADC internal reference voltage characteristics	33
5.3.19.	COMP internal reference voltage characteristics (4bit DAC)	34
5.3.20.	Timer characteristics	34
6.	Package information	36
6.1.	QFN20	36
6.2.	QFN16	37
7.	Ordering information	38
8.	Version history	39

1. Introduction

PY32C642 series microcontrollers are MCUs with high performance 32 bits ARM® Cortex®-M0+ core and wide voltage operating range. It has embedded 24 Kbytes Flash and 3 Kbytes SRAM memory, a maximum operating frequency of 24 MHz, and contains various products in different package types. The chip integrates one channel 12 bits ADC, two 16 bits timers, and two channel comparators.

PY32C642 series microcontrollers' working temperature ranges from -40 to 85 °C with operating voltage from 1.7 ~ 5.5 V. The chip provides sleep and stop low-power operating modes for different low-power applications.

PY32C642 series microcontrollers are suitable for various applications, such as controllers, portable devices, PC peripherals, gaming and GPS platforms, industrial applications.

Table 1-1 PY32C642 Series Product Planning and Features

Peripherals		PY32C642F15U6	PY32C642W15U6
Flash (Kbytes)		24	
SRAM (Kbytes)		3	
Timers	Advanced	1 (16-bit)	
	General purpose	1 (16-bit)	
	Low-power	1	
	SysTick	1	
	Watchdog	1	
GPIOs		18	14
12-bit ADC (external + internal)		8+2	
Comparators		2	
Max. CPU frequency		24 MHz	
Operating Temperature		-40 ~ 85 °C	
Operating Voltage		1.7 ~ 5.5 V	
Package		QFN20	QFN16

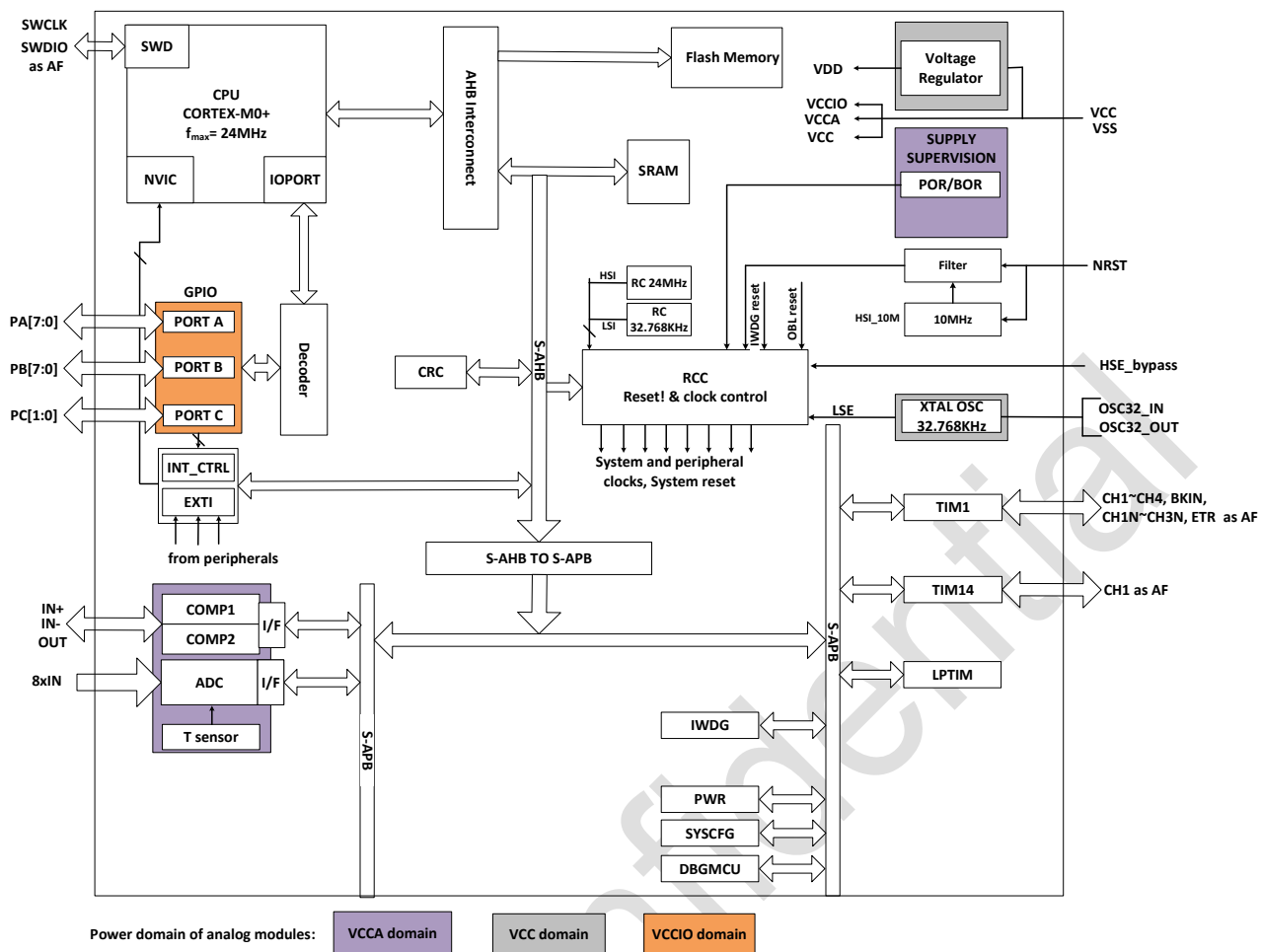


Figure 1-1 Functional Module

2. Functional overview

2.1. Arm® Cortex®-M0+ core

Arm® Cortex®- M0+ is an entry-level 32 bits Arm Cortex processor designed for a wide range of embedded applications. It provides developers with significant benefits, including:

- Simple structure, easy to learn and program
- Ultra-low power consumption, energy-saving operation
- Reduced code density and more

Cortex-M0+ processor has a 32 bits core, optimized for area and power consumption, as well with a 2-stage pipeline Von Neumann architecture. The processor offers high-end processing hardware, including single-cycle multipliers, through a streamlined but powerful instruction set and an extensively optimized design. Moreover, it delivers the superior performance expected from a 32 bits architecture computer, with a higher coding density than other 8 and 16 bits microcontrollers.

Cortex-M0+ is tightly coupled with a Nested Vectored Interrupt Controller(NVIC).

2.2. Memory

The on-chip integrated SRAM is accessed by bytes (8 bits), half-word (16 bits) or word (32 bits).

The on-chip integrated Flash consists of two different physical areas:

- Main flash area, which consists application and user data, in addition, you can set the maximum 4 Kbytes load flash to be used as the User bootloader.
- The information area has 768 Bytes, and it includes the following parts:
 - Option bytes
 - UID bytes
 - Factory config bytes
 - USER OTP memory

The protection mechanisms of Flash main memory includes the following ones:

- Write protection (WRP) control prevents unwanted writes (confuse by program memory pointer from PC). The minimum protection unit for write protection is 4 Kbytes.
- Option byte write protection has special unlocking design.

2.3. Boot mode

Through configuration bit nBOOT0/ nBOOT1(stored in Option bytes), three different boot modes can be selected, as shown in the following table:

Table 2-1 Boot configuration

Boot mode configuration		Mode	
nBOOT1 bit	nBOOT0 bit	Boot memory size ==0	Boot memory size !=0
X	0	Main flash boots	Main Flash boots
0	1	SRAM boots	SRAM boots
1	1	N/A	Load Flash boots

2.4. Clock System

After the CPU starts, the default system clock frequency is HSI 24 MHz, and the system clock frequency and system clock source can be reconfigured after program runs. The high frequency clocks than can be selected are:

- 24 MHz configurable internal high precision HIS clock
- A 32.768 KHz configurable LSI clock
- 4 MHz ~ 32 MHz external input clock
- 32.768 KHz LSE clock

The AHB clock can be divided based on the system clock, and the APB clock can be divided based on the AHB clock. AHB and APB clock frequencies are up to 24 MHz.

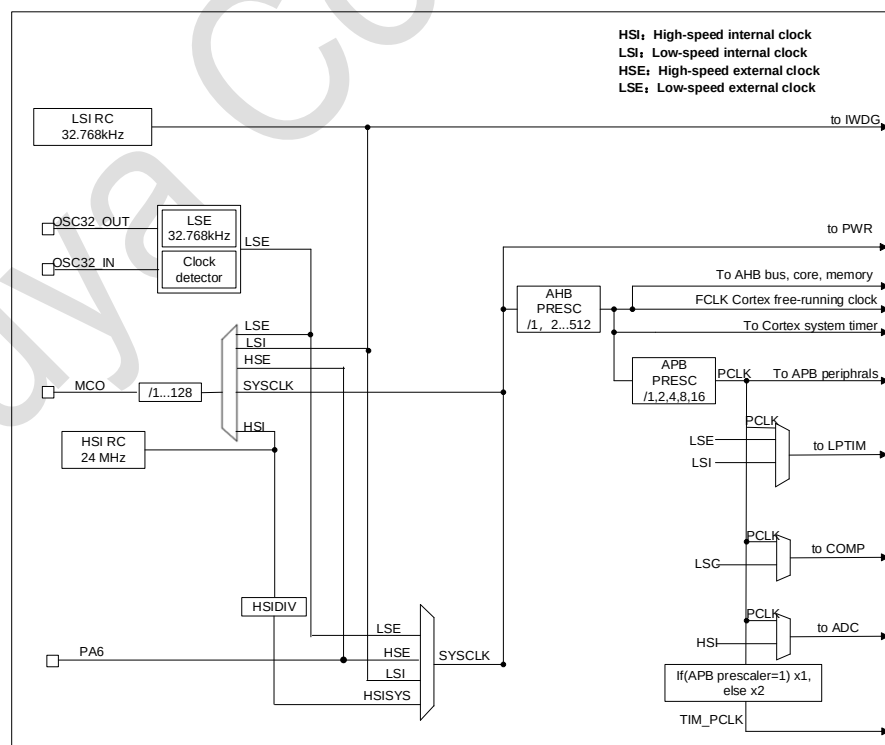


Figure 2-1 System Clock Structure Diagram

2.5. Power Management

2.5.1. Power block diagram

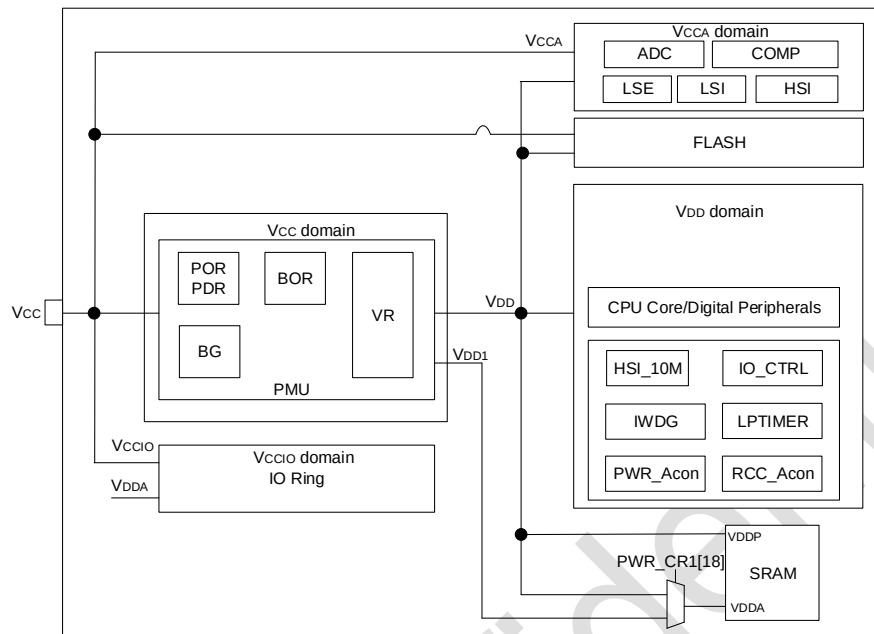


Figure 2-2 Power Block Diagram

Table 2-2 Power Block Diagram

No.	Power supply	Power value	Description
1	V _{CC}	1.7 V ~ 5.5 V	The chip is supplied with power through the power pins, and its power supply module is part of the analogue circuit.
2	V _{CCA}	1.7 V ~ 5.5 V	Power to most analogue modules from V _{CC} PAD (a separate power PAD can also be designed).
3	V _{CCIO}	1.7 V ~ 5.5 V	Supply power to IO, from V _{CC} PAD

2.5.2. Power monitoring

2.5.2.1. Power on reset/ Power down rest (POR/PDR)

The embedded Power on reset (POR)/Power down reset (PDR) modules are designed to provide power-on and power-off reset for the chip. The modules keep working in all modes.

2.5.2.2. Brown-out reset (BOR)

In addition to POR/PDR, BOR (brown out reset) has also been implemented. BOR can only be enabled and disabled through the option byte.

When the BOR is turned on, the BOR threshold can be selected by the Option byte, and both the rising and falling detection points can be configured individually.

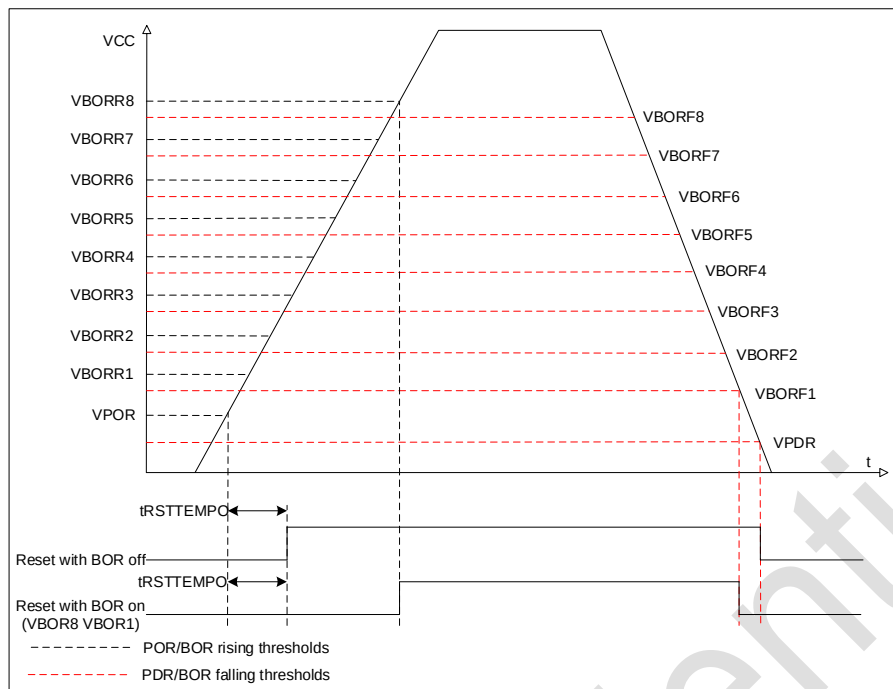


Figure 2-3 POR/PDR/BOR threshold

2.5.3. Voltage regulator

The chip designs two voltage regulators:

- MR (Main regulator) keeps working when the chip is in normal operating state
- LPR (low power regulator) provides a lower consumption option in stop mode

2.5.4. Low power mode

In addition to the normal operating mode, the chip has other two low-power modes:

- **Sleep mode:** Peripherals can be configured to keep working when the CPU clock is off (NVIC, SysTick, etc.). (It's recommended only to enable the modules that must work, and close the module after the module works.)
- **Stop mode:** In this mode, the contents of SRAM and registers are maintained, HIS is turned off. GPIO, IWDG, NRST, COMP output, LPTIM can wake up stop mode.

2.6. Rest

Two resets are designed in the chip: power and system reset.

2.6.1. Power reset

A power reset occurs in the following situations:

- Power on reset/ Power down reset (POR/PDR)
- Brown-out reset (BOR)

2.6.2. System reset

A system reset occurs when the following events occur:

- Reset of NRST pin
- Independent Watchdog Reset (IWDG)
- SYSRESETREQ software reset
- option byte load reset (OBL)
- Power reset (POR/PDR, BOR)

2.7. General purpose input and output (GPIO)

The software configures each GPIO as output (push-pull or open-drain), input (floating, pull-up/down, analogue), peripheral multiplexing function, and locking mechanism freeze I/O port configuration function.

2.8. Interrupt

The PY32C642 handles exceptions through the Cortex-M0+ processor's embedded Vectored Interrupt Controller (NVIC) and an Extended Interrupt/Event Controller (EXTI).

2.8.1. Interrupt controller NVIC

NVIC is a tightly coupled IP inside the Cortex-M0+ processor. The NVIC can handle NMI (Non-Maskable Interrupts) and maskable external interrupts from outside the processor and Cortex-M0+ internal exceptions. NVIC provides flexible priority management.

The tight coupling of the processor core to the NVIC greatly reduces the delay between an interrupt event and the initiation of the corresponding interrupt service routine (ISR). The ISR vectors are listed in a vector table, stored at a base address of the NVIC. The vector table base address determines the vector address of the ISR to execute, and the ISR is used as the offset composed of serial numbers.

If a high-priority interrupt event occurs and a low-priority interrupt event is just waiting to be serviced, the later arriving high-priority interrupt event will be serviced first. Another optimization is called tail-chaining. When returning from a high-priority ISR and then starting a pending low-priority ISR, unnecessary pushes and pops of processor contexts will be skipped. This reduces latency and improves power efficiency.

NVIC features:

- Low latency interrupt handling

- Level 4 Interrupt Priority
- Supports one NMI interrupt
- Supports 18 maskable external interrupts
- Supports 10 Cortex-M0+ exceptions
- High-priority interrupts can interrupt low-priority interrupt responses
- Supports tail-chaining optimization
- Hardware Interrupt Vector Retrieval

2.8.2. External interrupt/event controller (EXTI)

EXTI adds flexibility to handle physical wire events and generates wake-up events when the processor wakes up from stop mode.

The EXTI controller has multiple channels, including a maximum of 18 GPIOs, 2 COMP outputs, and LPTIM wake up signal. GPIO, COMP can be configured to be triggered by a rising edge, falling edge or double edge. Any GPIO signal can be configured as EXTI0 ~ 7 channel through the select signal.

Each EXTI line can be independently masked through registers.

The EXTI controller can capture pulses shorter than the internal clock period.

Registers in the EXTI controller latch each event. Even in stop mode, after the processor wakes up from stop mode, it can identify the wake-up source or identify the GPIO and event that caused the interrupt.

2.9. Analog to digital converter (ADC)

The chip has a 12 bits SAR ADC. The module has up to 10 channels, including 8 external channels and 2 internal channels. Reference voltage can be selected from on-chip precise voltage (1.5V, 2.048V, 2.5V) or V_{CC} power supply.

The conversion mode of each channel can be set to single, continuous, sweep, discontinuous mode. Conversion results are stored in left or right-aligned 16 bits data registers.

An analogue watchdog allows the application to detect if the input voltage exceeds a user-defined high or low threshold.

The ADC has been implemented to operate at a low frequency, resulting in lower power consumption.

At the end of sampling, conversion, and continuous conversion, an interrupt request is generated when the conversion voltage exceeds the threshold when simulating the watchdog.

2.10. Comparator (COMP)

The on-chip general purpose comparators (COMP) can also be used in combination with timers. The comparators can be used as follows:

- Wakeup from low-power mode triggered by an analog signal
- Analog signals conditioning
- Cycle-by-cycle current control loop when combined with a PWM output from a timer

2.10.1. COMP main features

- Each comparator has configurable positive and negative inputs used for flexible voltage selection:
 - Multiplexing I/O pin
 - Power supply V_{CC} and 15 submultiple values (1/16, 2/16 ... 15/16) provided by voltage divider
 - Reference voltage can be selected from on-chip precise voltage (1.5V, 2.048V, 2.5V) or V_{CC} power supply, and 15 submultiple values (1/16, 2/16 ... 15/16) provided by voltage divider
- The outputs can be redirected to an I/O or to timer inputs for triggering:
 - OCREF_CLR event (cycle by cycle current control)
 - Brakes for fast PWM shutdown

Each COMP has interrupt generation capability which is used to wake up the chip from low power modes (sleep and stop mode) (Via EXTI).

2.11. Timer

The characteristics of PY32C642 are shown in the following table:

Table 2-3 Timer characteristics

Types	Timer	Bit Width	Counting Direction	Prescaler	Capture/compare channel	Complementary output
Advanced	TIM1	16 bits	Superior, Down, Center aligned	1~65536	4	3
General purpose	TIM14	16 bits	Superior	1~65536	1	-

2.11.1. Advanced Timer

The advanced timer (TIM1) consists of a 16 bits auto-reload counter driven by a 16 bits programmable prescaler. It can be used in various scenarios, including pulse length measurement of input signals

(input capture) or generating output waveforms (output compare, output PWM, complementary PWM with dead-time insertion).

TIM1 includes 4 independent channels:

- Input capture
- Output comparison
- PWM generation (edge or center-aligned mode)
- Single pulse mode output

If TIM1 is configured as a standard 16-bit timer, it has the same characteristics as the TIMx timer. Full modulation capability (0-100%) if configured as a 16 bits PWM generator.

In the MCU debug mode, TIM1 can freeze counting.

The timer feature with the same architecture is shared so that the TIM1 can work with other timers for synchronization or event chaining through the timer chaining function.

2.11.2. General purpose timer

- The general purpose timer TIM14 consists of 16 bits programmable prescaler
- TIM14 has an independent channel for input capture/ output compare, PWM or single pulse mode output
- In the MCU debug mode, the TIM14 can freeze counting

2.11.3. Lower power timer

- LPTIM is a 16 -bit up counter with a 3 bits prescaler and only support a single count
- LPTIM can be configured as a stop mode wakeup source
- In the MCU debug mode, LPTIM can freeze the count value

2.11.4. IWDG

Independent watchdog (IWDG) is integrated in the chip, and this module has the characteristics of high-security level, accurate timing and flexible use. IWDG finds and resolves functional confusion due to software failure and triggers a system reset when the counter reaches the specified timeout value.

- The IWDG is clocked by LSI, so even if the main clock fails, it can keep working
- IWDG is the best suited for applications that require the watchdog as a standalone process outside of the main application and do not have high timing accuracy constraints.
- Controlling of option byte can enable IWDG hardware mode
- IWDG is the wake-up source of stop mode, which wakes up stop mode by reset

- In the MCU debug mode, IWDG can freeze the count value

2.11.5. SysTick timer

SysTick counters are specifically for real-time operating systems (RTOS) also can use as standard down counters.

SysTick Features:

- 24 bits count down
- Self-loading capability
- An interrupt can be generated when the counter reaches 0 (maskable)

2.12. SWD

The ARM SWD interface allows serial debugging tools to be connected to the PY32C642.

3. Pin configuration

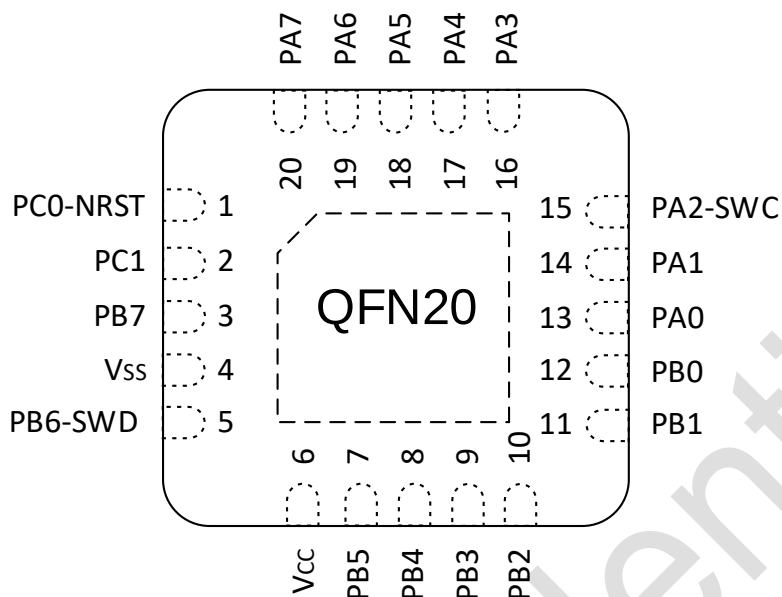


Figure 3-1 QFN20 Pinout1 PY32C642F15U6 (Top view)

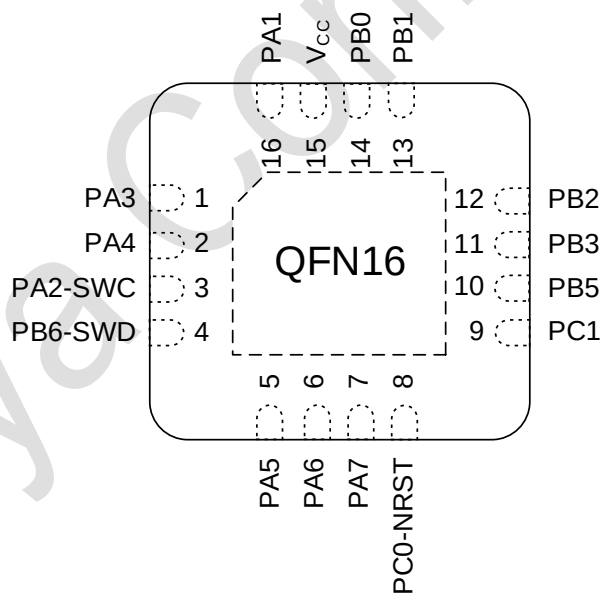


Figure 3-2 QFN16 Pinout1 PY32C642W15U6 (Top view)

Table 3-1 Pin definition terminology and symbols

Type		Symbol	Definition
Port type		S	Supply pin
		G	Ground pin
		I/O	Input/output pin
		NC	Undefined
Port structure		COM	5 V port, with internal input and output function
		RST	-
Notes		-	Unless other specified, all ports are used as analog inputs between and after reset
Port function	Multiplexing function	-	Function selected by GPIOx_AFR register
	Additional features	-	Directly selected or enabled through peripheral registers

Table 3-2 Pin definition

Package type		Reset	Port type	Port structure	Port function	
QFN16 W1	QFN20 F1				Multiplexing function	Additional features
5	18	PA5	I/O	COM	TIM1_CH1 TIM14_CH1	
6	19	PA6	I/O	COM	EVENTOUT	ADC_IN3 External_clock_in
7	20	PA7	I/O	COM	TIM1_CH4 MCO	ADC_IN4
8	1	PC0-NRST ⁽¹⁾ ⁽³⁾	I/O	RST	SWDIO TIM1_CH1N EVENTOUT	NRST ADC_IN5
9	2	PC1-OSCIN	I/O	COM	-	OSCIN
-	3	PB7-OSCOUT	I/O	COM	TIM14_CH1	OSCOUT
-	4	V _{SS}	S		Ground	
4	5	PB6(SWDIO) ⁽²⁾	I/O	COM	SWDIO	ADC_IN6
15	6	V _{CC}	S		Digital power supply	
10	7	PB5	I/O	COM	TIM1_CH3 TIM14_CH1	
-	8	PB4	I/O	COM	TIM1_BKIN	
11	9	PB3	I/O	COM	TIM1_ETR	

Package type		Reset	Port type	Port structure	Port function	
QFN16 W1	QFN20 F1				Multiplexing function	Additional features
					CMP1_OUT	
12	10	PB2	I/O	COM	TIM1_CH1N	
					TIM1_CH3	
13	11	PB1	I/O	COM	TIM1_CH2N	ADC_IN0 CMP1_INP CMP1_INM
					TIM1_CH4	
					MCO	
14	12	PB0	I/O	COM	TIM1_CH2	ADC_IN7 CMP1_INM
					TIM1_CH3N	
-	13	PA0	I/O	COM	TIM1_CH1	
16	14	PA1	I/O	COM	TIM1_CH2	
3	15	PA2(SWCLK) ⁽²⁾ ⁽³⁾	I/O	COM	SWCLK	
					TIM1_CH4	
					CMP2_OUT	
1	16	PA3	I/O	COM	TIM1_CH2	ADC_IN1 CMP2_INP CMP2_INM
2	17	PA4	I/O	COM	TIM1_CH3	ADC_IN2 CMP2_INM
					TIM14_CH1	

1. Selecting PC0 or NRST/SWDIO is configured through option bytes.
2. After reset (when option byte configures 0/0,0/1,1/0), the two pins of PB6 and PA2 are configured as SWDIO and SWCLK AF function, the former internal pull-up resistor, the latter pull-down resistor is activated.
3. After reset (when option byte configures 1/1), the two pins of PC0 and PA2 are configured as SWDIO and SWCLK AF function, the former internal pull-up resistor, the latter pull-down resistor is activated.

3.1. Port A multiplexing function mapping

Table 3-3 Port multiplexing function mapping

Port	AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7
PA0	-	-	TIM1_CH1	-	-	-	-	-
PA1	-	-	TIM1_CH2	-	-	-	-	-
PA2	SWC	-	TIM1_CH4	-	CMP2_OUT	-	-	-
PA3	-	-	TIM1_CH2	-	-	-	-	-
PA4	-	-	TIM1_CH3	-	-	TIM14_CH1	-	-
PA5	-	-	TIM1_CH1	-	-	TIM14_CH1	-	-
PA6	-	-	-	-	-	-	-	EVENTOUT
PA7	-	-	TIM1_CH4	-	MCO	-	-	-

3.2. Port B multiplexing function mapping

Table 3-4 Port B multiplexing function mapping

Port	AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7
PB0	-	-	TIM1_CH2	TIM1_CH3N	-	-	-	-
PB1	-	-	TIM1_CH2N	TIM1_CH4	MCO	-	-	-
PB2	-	-	TIM1_CH1N	TIM1_CH3	-	-	-	-
PB3	-	-	TIM1_ETR	-	CMP1_OUT	-	-	-
PB4	-	-	TIM1_BKIN	-	-	-	-	-
PB5	-	-	TIM1_CH3	-	-	TIM14_CH1	-	-
PB6	SWD	-	-	-	-	-	-	-
PB7	-	-	-	-	-	TIM14_CH1	-	-

3.3. Port C multiplexing function mapping

Table 3-5 multiplexing function mapping

Port	AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7
PC0	SWD	-	TIM1_CH1N	-	-	-	-	EVENTOUT
PC1	-	-	-	-	-	-	-	-

4. Memory map

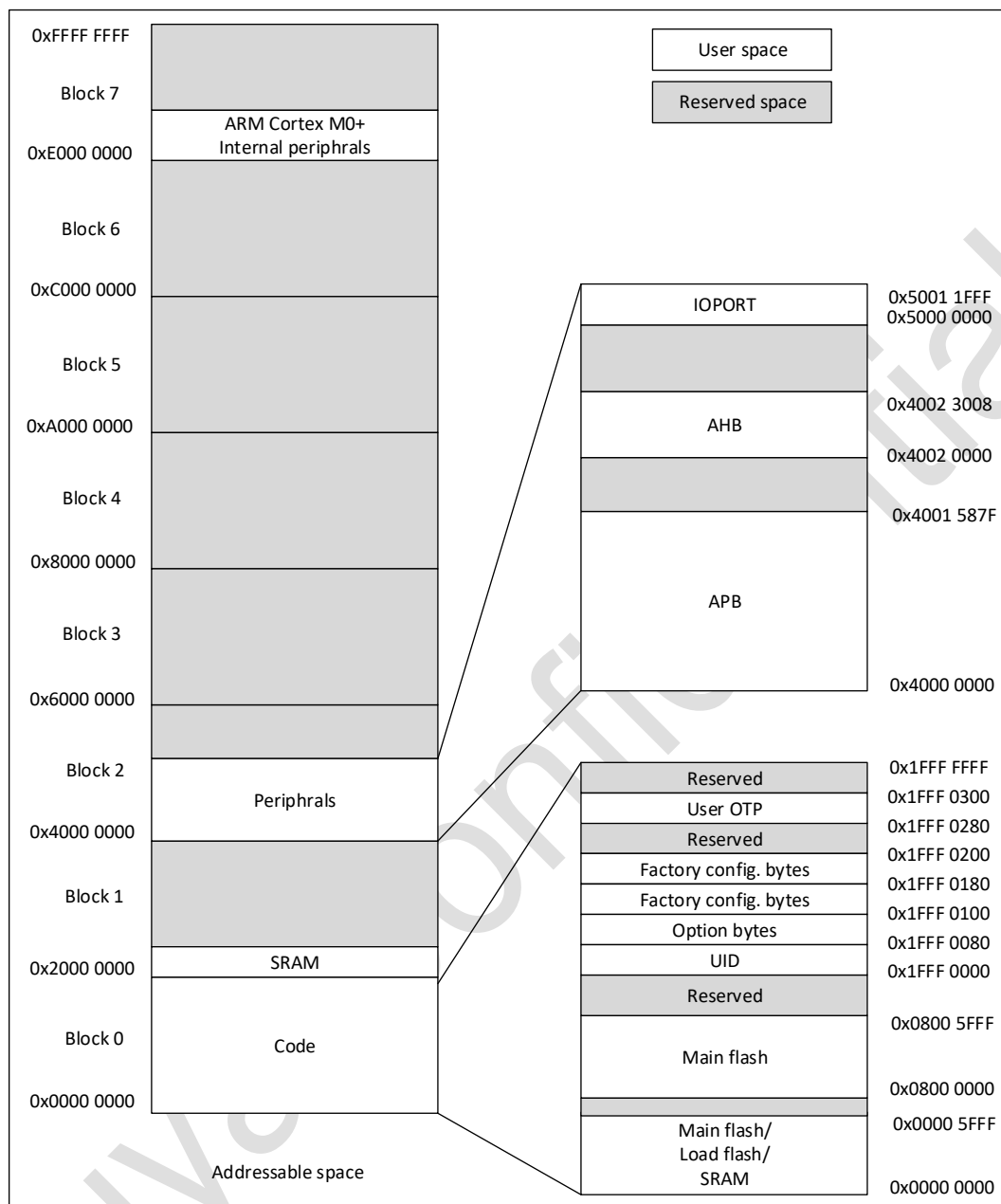


Figure 4-1 Memory map

Table 4-1 Memory address

Type	Boundary Address	Size	Memory Area
SRAM	0x2000 C000-0x3FFF FFFF	-	Reserved
	0x2000 0000-0x2000 0BFF	3 Kbytes	SRAM
Code	0x1FFF 0300-0x1FFF FFFF	-	Reserved
	0x1FFF 0280-0x1FFF 02FF	128 Bytes	USER OTP memory
	0x1FFF 0180-0x1FFF 01FF	128 Bytes	Factory Configuration bytes
	0x1FFF 0100-0x1FFF 017F	128 Bytes	Factory Configuration bytes
	0x1FFF 0080-0x1FFF 00FF	128 Bytes	Option bytes
	0x1FFF 0000-0x1FFF 007F	128 Bytes	UID
	0x0800 6000-0x1FFE FFFF	-	Reserved
	0x0800 0000-0x0800 5FFF	24 Kbytes	Main flash memory
	0x0000 6000-0x07FF FFFF	-	Reserved
	0x0000 0000-0x0000 5FFF	24 Kbytes	Select based on Boot configuration: 1) Main flash memory 2) Load flash 3) SRAM

Table 4-2 Peripheral register address

Bus	Boundary Address	Size	Peripheral
	0xE000 0000-0xE00F FFFF	-	M0+
IOPORT	0x5000 0C00-0x5FFF FFFF	-	Reserved
	0x5000 0800-0x5000 0BFF	1 Kbytes	GPIOC
	0x5000 0400-0x5000 07FF	1 Kbytes	GPIOB
	0x5000 0000-0x5000 03FF	1 Kbytes	GPIOA
AHB	0x4002 3400-0x4FFF FFFF	-	Reserved
	0x4002 300C-0x4002 33FF	1 Kbytes	Reserved
	0x4002 3000-0x4002 3008		CRC
	0x4002 2400-0x4002 2FFF	-	Reserved
	0x4002 2000-0x4002 23FF	1 Kbytes	Flash
	0x4002 1C00-0x4002 1FFF	-	Reserved
	0x4002 1900-0x4002 1BFF	1 Kbytes	Reserved
	0x4002 1800-0x4002 18FF		EXTI
	0x4002 1400-0x4002 17FF	-	Reserved
	0x4002 1080-0x4002 13FF	1 Kbytes	Reserved
	0x4002 1000-0x4002 107F		RCC
	0x4002 0000-0x4002 0FFF	-	Reserved

Bus	Boundary Address	Size	Peripheral
APB	0x4001 5C00-0x4001 FFFF	-	Reserved
	0x4001 5880-0x4001 5BFF	1 Kbytes	Reserved
	0x4001 5800-0x4001 587F		DBG
	0x4001 3C00-0x4001 57FF	-	Reserved
	0x4001 381C-0x4001 3BFF	1 Kbytes	Reserved
	0x4001 3800-0x4001 3018		
	0x4001 3400-0x4001 37FF	-	Reserved
	0x4001 3010-0x4001 33FF	1 Kbytes	Reserved
	0x4001 3000-0x4001 300C		
	0x4001 2C50-0x4001 2FFF	1 Kbytes	Reserved
	0x4001 2C00-0x4001 2C4C		TIM1
	0x4001 2800-0x4001 2BFF	-	Reserved
	0x4001 270C-0x4001 27FF	1 Kbytes	Reserved
	0x4001 2400-0x4001 2708		ADC
	0x4001 0400-0x4001 23FF	-	Reserved
	0x4001 0220-0x4001 03FF	1 Kbytes	Reserved
	0x4001 0200-0x4001 021F		COMP1/2
	0x4001 0000-0x4001 01FF		SYSCFG
	0x4000 8000-0x4000 FFFF	-	Reserved
	0x4000 7C28-0x4000 7FFF	1 Kbytes	Reserved
	0x4000 7C00-0x4000 7C24		LPTIM
	0x4000 7400-0x4000 7BFF	-	Reserved
	0x4000 7018-0x4000 73FF	1 Kbytes	Reserved
	0x4000 7000-0x4000 7014		PWR
	0x4000 5800-0x4000 6FFF	-	Reserved
	0x4000 5434-0x4000 57FF	1 Kbytes	Reserved
	0x4000 5400-0x4000 5430		
	0x4000 3400-0x4000 53FF	-	Reserved
	0x4000 3014-0x4000 33FF	1 Kbytes	Reserved
	0x4000 3000-0x4000 0010		IWDG
	0x4000 2400-0x4000 2FFF	-	Reserved
	0x4000 2054-0x4000 23FF	1 Kbytes	Reserved
	0x4000 2000-0x4000 0050		TIM14
	0x4000 0000-0x4000 1FFF	-	Reserved

5. Electrical characteristics

5.1. Test conditions

Unless otherwise specified, all voltages are referenced to V_{SS} .

5.1.1. Min and Max

Unless otherwise specified, the chip is screened by mass production testing at ambient temperature $T_A = 25\text{ }^{\circ}\text{C}$ and $T_A = T_A(\text{max})$, guaranteed to reach the minimum value and maximum value under the worst ambient temperature, supply voltage and clock frequency conditions.

Based on electrical characterization results, design simulations, and/or process parameters noted below the table, not tested in production. Minimum and maximum values are referenced to sample testing and averaged plus or minus three times the standard deviation.

5.1.2. Typical value

Unless otherwise specified, typical data is based on $T_A = 25\text{ }^{\circ}\text{C}$ and $V_{CC} = 3.3\text{ V}$. These data are for design guidance only and have not been tested.

Typical ADC accuracy values are obtained by sampling a standard batch, tested under all temperature ranges, and 95 % of the chip error is less than or equal to the given value.

5.2. Absolute maximum ratings

If the applied voltage exceeds the absolute maximum value given in the table below, it may cause permanent damage to the chip. Only the strength ratings that can be tolerated are listed here, and it does not imply that the functional operation of the device is correct under these conditions. Operating under maximum conditions for a long time may affect the reliability of the chip.

Table 5-1 Voltage characteristics⁽¹⁾

Symbol	Descriptions	Minimum	Maximum	Unit
V_{CC}	External mains power supply	-0.3	6.25	V
V_{IN}	Input voltage of other pins	-0.3	$V_{CC}+0.3$	V

1. Power supply V_{CC} and ground V_{SS} pins must always be connected to the external power supply within the allowable range.

Table 5-2 Current characteristics

Symbol	Descriptions	Maximum	Unit
I_{VCC}	Total current flowing into V_{CC} pin supply current ⁽¹⁾	80	mA
I_{VSS}	Total current flowing out of V_{SS} pin (outflow current) ⁽¹⁾	80	mA
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin	20	mA
	Output current source by any I/Os and control pin	-20	

1. Power supply V_{CC} and ground V_{SS} pins must always be connected to the external power supply within the allowable range.

Table 5-3 Temperature characteristics

Symbol	Descriptions	Value	Unit
T_{STG}	Storage temperature	-65 ~ +150	°C
T_O	Operating temperature	-40 ~ +85	°C

5.3. Operating conditions

5.3.1. General operating conditions

Table 5-4 General operating conditions

Symbol	Parameter	Conditions	Minimum	Maximum	Unit
f_{HCLK}	Internal AHB clock frequency	-	0	24	MHz
f_{PCLK}	Internal APB clock frequency	-	0	24	MHz
V_{CC}	Standard operating voltage	-	1.7	5.5	V
V_{IN}	IO input voltage	-	-0.3	$V_{CC} + 0.3$	V
T_A	Ambient temperature	-	-40	85	°C
T_J	Junction temperature	-	-40	90	°C

5.3.2. Power on and down operating conditions

Table 5-5 Power on and Power down operation conditions

Symbol	Parameter	Conditions	Minimum	Maximum	Unit
t_{VCC}	V_{CC} rise rate	-	0	∞	$\mu s/V$
	V_{CC} fall rate	-	20	∞	

5.3.3. Embedded reset module characteristics

Table 5-6 Embedded reset module characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
$t_{RSTEMPO}^{(1)}$	Reset time	-	-	4.0	7.5	ms
$V_{POR/PDR}$	POR/PDR reset threshold	Rising edge	1.5 ⁽²⁾	1.6	1.7	V
		Falling edge	1.45	1.55	1.65 ⁽²⁾	V
$V_{PDRhyst}^{(1)}$	PDR hysteresis	-	-	50	-	mV
V_{BOR}	BOR threshold voltage	BOR_LEV[2:0]=000 (Rising edge)	1.7 ⁽²⁾	1.8	1.9	V
		BOR_LEV[2:0]=000 (Falling edge)	1.6	1.7	1.8 ⁽²⁾	V
		BOR_LEV[2:0]=001 (Rising edge)	1.9 ⁽²⁾	2	2.1	V
		BOR_LEV[2:0]=001 (Falling edge)	1.8	1.9	2 ⁽²⁾	V

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
		BOR_LEV[2:0]=010 (Rising edge)	2.1 ⁽²⁾	2.2	2.3	V
		BOR_LEV[2:0]=010 (Falling edge)	2	2.1	2.2 ⁽²⁾	V
		BOR_LEV[2:0]=011 (Rising edge)	2.3 ⁽²⁾	2.4	2.5	V
		BOR_LEV[2:0]=011 (Falling edge)	2.2	2.3	2.4 ⁽²⁾	V
		BOR_LEV[2:0]=100 (Rising edge)	2.5 ⁽²⁾	2.6	2.7	V
		BOR_LEV[2:0]=100 (Falling edge)	2.4	2.5	2.6 ⁽²⁾	V
		BOR_LEV[2:0]=101 (Rising edge)	2.7 ⁽²⁾	2.8	2.9	V
		BOR_LEV[2:0]=101 (Falling edge)	2.6	2.7	2.8 ⁽²⁾	V
		BOR_LEV[2:0]=110 (Rising edge)	2.9 ⁽²⁾	3	3.1	V
		BOR_LEV[2:0]=110 (Falling edge)	2.8	2.9	3 ⁽²⁾	V
		BOR_LEV[2:0]=111 (Rising edge)	3.1 ⁽²⁾	3.2	3.3	V
		BOR_LEV[2:0]=111 (Falling edge)	3	3.1	3.2 ⁽²⁾	V
V_BOR_hyst	BOR hysteresis	-	-	100	-	mV

1. Guaranteed by design, not tested in production.
2. Data is based on assessment results and is not tested in production.

5.3.4. Operating current characteristics

Table 5-7 Current consumption in run mode

Symbol	Conditions						Typical (1)	Maximum	Unit
	System clock	Frequency	Code	Run	Peripheral clock	FLASH sleep			
I _{DD} (run)	HSI	24 MHz	While ⁽¹⁾	Flash	ON	DISABLE	1.10	-	mA
					OFF	DISABLE	0.90	-	
	LSI	32.768 kHz			ON	DISABLE	160	-	μA
					OFF	DISABLE	160	-	
		32.768 kHz			ON	ENABLE	108	-	
					OFF	ENABLE	108	-	

1. Data is based on assessment results and is not tested in production.

Table 5-8 Current consumption in sleep mode

Symbol	Conditions				Typical ⁽¹⁾	Maximum	Unit
	System clock	Frequency	Peripheral clock	FLASH sleep			
I _{DD} (sleep)	HSI	24 MHz	ON	DISABLE	0.80	-	mA
			OFF	DISABLE	0.50	-	
	LSI	32.768 kHz	ON	DISABLE	159.3	-	μA

Symbol	Conditions				Typical ⁽¹⁾	Maximum	Unit
	System clock	Frequency	Peripheral clock	FLASH sleep			
		32.768 kHz	OFF	DISABLE	158.9	-	
			ON	ENABLE	85.3	-	
			OFF	ENABLE	84.8	-	

1. Data is based on assessment results and is not tested in production.

Table 5-9 Current consumption in stop mode

Symbol	Conditions				Typical ⁽¹⁾	Maximum	Unit
	V _{CC}	MR/LPR	LSI	Peripheral clock			
I _{DD} (stop)	1.7 ~ 5.5 V	MR	-	-	75.3	-	μA
		LPR	ON	IWDG+LPTIM	1.70	-	
				IWDG	1.70	-	
				LPTIM	1.70	-	
			OFF	No	1.50	-	

1. Data is based on assessment results and is not tested in production.

5.3.5. Wakeup time from low-power modes

Table 5-10 Low power mode wake-up time

Symbol	Parameter ⁽¹⁾		Conditions	Typical ⁽²⁾	Maximum	unit
t _{WUSLEEP}	Wake-up time from sleep		-	0.6	-	μs
t _{WUSTOP}	Wake-up time from stop	Powered by MR	Execute program in Flash, HSI(24 Mhz)as system clock	6.4	-	
		Powered by LPR	Execute program in Flash, HIS as system clock(24 M)	10.6	-	

1. The wake-up time is measured from the wake-up time until the first instruction is read by the user program.
2. Data is based on assessment results and is not tested in production.

5.3.6. External clock source characteristics

5.3.6.1. External high-speed clock

In the bypass mode of HSE (the HSEBYP of RCC_CR is set), when the high-speed start-up circuit in the chip stops working, the corresponding IO is used as a standard GPIO.

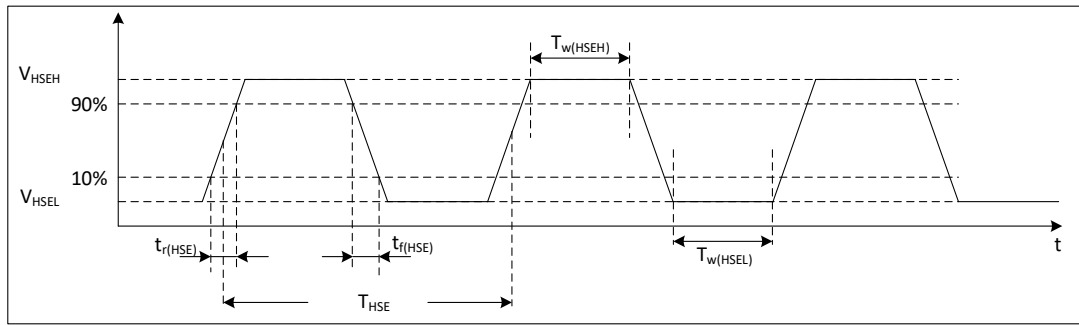


Figure 5-1 External high-speed clock timing diagram

Table 5-11 External high-speed clock features

Symbol	Parameter ⁽¹⁾	Minimum	Typical	Maximum	Unit
f_{HSE_ext}	User external clock frequency	1	8	32	MHz
V_{HSEH}	Input pin high level voltage	$0.7 V_{CC}$	-	V_{CC}	V
V_{HSEL}	Input pin low level voltage	V_{SS}	-	$0.3 V_{CC}$	
$t_{w(HSEH)}$ $t_{w(HSEL)}$	Enter high or low time	15	-	-	ns
$t_{r(HSE)}$ $t_{f(HSE)}$	Enter the rise/fall time	-	-	20	ns

1. Guaranteed by design, not tested in production.

5.3.6.2. External low-speed clock

In the bypass mode of LSE (the LSEBYP of RCC_BDCR is set), when the low-speed start-up circuit in the chip stops working, the corresponding IO is used as a standard GPIO.

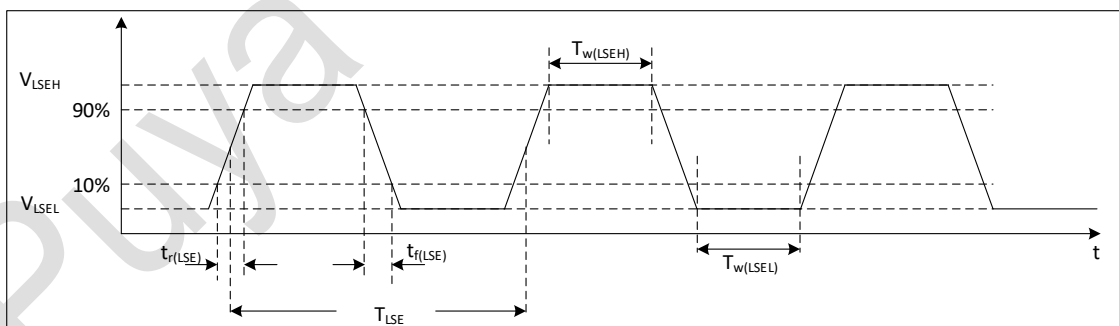


Figure 5-2 External low-speed clock timing diagram

Table 5-12 External low-speed clock features

Symbol	Parameter ⁽¹⁾	Minimum	Typical	Maximum	Unit
f_{LSE_ext}	User external clock frequency	-	32.768	1000	KHz
V_{LSEH}	Input pin high level voltage	$0.7 * V_{CC}$	-	-	V
V_{LSEL}	Input pin low level voltage	-	-	$0.3 * V_{CC}$	V

Symbol	Parameter ⁽¹⁾	Minimum	Typical	Maximum	Unit
$t_{W(LSEH)}$ $t_{W(LSEL)}$	Enter high or low time	450	-	-	ns
$t_{r(LSE)}$ $t_{f(LSE)}$	Enter the rise or fall time	-	-	50	ns

1. Guaranteed by design, not tested in production.

5.3.6.3. External low-speed crystal

An external 32.768 KHz crystal/ceramic resonator. In the application, the crystal and load capacitors should be as close as possible to the pins to minimize output distortion and start-up settling time.

Table 5-13 External low-speed crystal characteristics

Symbol	Parameter	Conditions ⁽¹⁾	Minimum	Typical	Maximum	Unit
$I_{DD}^{(4)}$	LSE power consumption	LSE_DRIVER [1:0] = 00	-	100	-	nA
		LSE_DRIVER [1:0] = 01	-	700	-	
		LSE_DRIVER [1:0] = 10	-	1200	-	
		LSE_DRIVER [1:0] = 11	-	1600	-	
$t_{SU(LSE)}^{(3) (4)}$	Start Time	-	-	3	-	s

1. Crystal/ceramic resonator characteristics are based on the manufacturer datasheet.
2. Guaranteed by design, not tested in production.
3. $t_{SU(LSE)}$ is the start-up time from enable (by software) to the clock oscillation reaches stability, measured for a standard crystal/resonator, which can vary greatly from one crystal/resonator to another.
4. Data is based on assessment results and is not tested in production.

5.3.7. Internal high frequency clock source HSI characteristics

Table 5-14 Internal high frequency clock source characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
f_{HSI}	HSI frequency	$T_A = 25^{\circ}\text{C}, V_{CC} = 3.3\text{ V}$	23.83 ⁽²⁾	24	24.17 ⁽²⁾	MHz
$\Delta_{Temp(HSI)}$	HSI frequency temperature drift 24 MHz	$V_{CC} = 2.0 \sim 5.5\text{ V}$ $T_A = -40 \sim 85^{\circ}\text{C}$	-2 ⁽²⁾	-	2 ⁽²⁾	%
		$V_{CC} = 1.7 \sim 5.5\text{ V}$ $T_A = 0 \sim 85^{\circ}\text{C}$	-2 ⁽²⁾	-	2 ⁽²⁾	
		$V_{CC} = 1.7 \sim 5.5\text{ V}$ $T_A = -40 \sim 85^{\circ}\text{C}$	-4 ⁽²⁾	-	2 ⁽²⁾	
$f_{TRIM}^{(1)}$	HSI fine-tuning accuracy	-	-	0.1	-	%

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
D _{HSI} ⁽¹⁾	Duty cycle	-	45	-	55	%
t _{Stab(HSI)}	HSI stabilization time	-	-	2	4 ⁽¹⁾	μs
I _{DD(HSI)} ⁽²⁾	HSI power consumption	24 MHz	-	193	-	μA

1. Guaranteed by design, not tested in production.
2. Data is based on assessment results and is not tested in production.

5.3.8. Internal low frequency clock source LSI characteristics

Table 5-15 Internal low frequency clock characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
f _{LSI}	LSI frequency	T _A = 25 °C, V _{CC} = 3.3 V	31.6	-	33.6	KHz
Δ _{Temp(LSI)}	LSI frequency temperature drift	V _{CC} = 1.7 ~ 5.5 V T _A = 0 ~ 85 °C	-10 ⁽²⁾	-	10 ⁽²⁾	%
		V _{CC} = 1.7 ~ 5.5 V T _A = -40 ~ 85 °C	-20 ⁽²⁾	-	20 ⁽²⁾	
f _{TRIM} ⁽¹⁾	LSI fine-tuning accuracy	-	-	0.2	-	%
t _{Stab(LSI)} ⁽¹⁾	LSI stabilization time	-	-	150	-	μs
I _{DD(LSI)} ⁽¹⁾	LSI power consumption	-	-	210	-	nA

1. Guaranteed by design, not tested in production.
2. Data is based on assessment results and is not tested in production.

5.3.9. Memory characteristics

Table 5-16 Memory characteristics

Symbol	Parameter	Conditions	Typical	Maximum ⁽¹⁾	Unit
t _{prog}	Page program	-	1.0	1.5	ms
t _{ERASE}	Page/sector/mass erase	-	3.5	5.0	ms
I _{DD}	Page program	-	2.1	2.9	mA
	Page/sector/mass erase	-	2.1	2.9	

1. Guaranteed by design, not tested in production.

Table 5-17 Memory erase times and data retention

Symbol	Parameter	Conditions	Minimum ⁽¹⁾	Unit
N _{END}	Erase and write time	T _A = -40 ~ 85 °C	100	Kcycle

Symbol	Parameter	Conditions	Minimum ⁽¹⁾	Unit
t _{RET}	Date retention period	10 Kcycle T _A = 55 °C	20	Year

1. Data is based on assessment results and is not tested in production.

5.3.10. EFT characteristics

Table 5-18 EFT characteristics

Symbol	Parameter	Conditions	Grade	Typical	Unit
EFT to Power	-	IEC61000-4-4	A	4	KV

5.3.11. ESD & LU characteristics

Table 5-19 ESD & LU characteristics

Symbol	Parameter	Conditions	Typical	Unit
V _{ESD(HBM)}	Static Discharge Voltage(human body model)	ESDA/JEDEC JS-001-2017	6	KV
V _{ESD(CDM)}	Static Discharge Voltage(charging equipment model)	ESDA/JEDEC JS-002-2018	1	KV
V _{ESD(MM)}	Static discharge voltage(machine model)	JESD22-A115C	200	V
LU	Static Latch-Up	JESD78E	200	mA

5.3.12. Port characteristics

Table 5-20 IO static characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V _{IH}	Input high level voltage	V _{CC} = 1.7 ~ 5.5 V	0.7 * V _{CC}	-	-	V
V _{IL}	Input low level voltage	V _{CC} = 1.7 ~ 5.5 V	-	-	0.3 * V _{CC}	V
V _{hys} ⁽¹⁾	Schmitt hysteresis voltage	-	-	200	-	mV
I _{lkg}	Input leakage current	-	-	-	1	μA
R _{PU}	Pull-up resistor	-	30	50	70	KΩ
R _{PD}	Pull-down resistor	-	30	50	70	KΩ
C _{IO} ⁽¹⁾	Pin capacitance	-	-	5	-	pF

1. Guaranteed by design, not tested in production.

Table 5-21 Output Voltage Characteristics

Symbol	Parameter ⁽¹⁾	Conditions		Minimum	Maximum	Unit
$V_{OL}^{(2)}$	COM IO output low level	GPIOx_OSPEEDR =11	$I_{OL} = 20\text{ mA},$ $V_{CC} \geq 5.0\text{ V}$	-	0.4	V
			$I_{OL} = 8\text{ mA},$ $V_{CC} \geq 2.7\text{ V}$	-	0.4	
			$I_{OL} = 4\text{ mA},$ $V_{CC} = 1.8\text{ V}$	-	0.5	
$V_{OH}^{(2)}$	COM IO output high level	GPIOx_OSPEEDR =11	$I_{OH} = 18\text{ mA},$ $V_{CC} \geq 5.0\text{ V}$	$V_{CC}-0.6$	-	V
			$I_{OH} = 8\text{ mA},$ $V_{CC} \geq 2.7\text{ V}$	$V_{CC}-0.4$	-	
			$I_{OH} = 4\text{ mA},$ $V_{CC} = 1.8\text{ V}$	$V_{CC}-0.5$	-	

- IO types can refer to the terms and symbols defined by the pins.
- Guaranteed by design, not tested in production.

5.3.13. NRST pin characteristics

Table 5-22 NRST pin characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V_{IH}	Input high level voltage	$V_{CC} = 1.7 \sim 5.5\text{ V}$	$0.7 * V_{CC}$	-	-	V
V_{IL}	Input low level voltage	$V_{CC} = 1.7 \sim 5.5\text{ V}$	-	-	$0.2 * V_{CC}$	V
$V_{hys}^{(1)}$	Schmitt hysteresis voltage	-	-	300	-	mV
I_{lkg}	Input leakage current	-	-	-	1	μA
$R_{PU}^{(1)}$	Pull-up resistor	-	30	50	70	$k\Omega$
$R_{PD}^{(1)}$	Pull-down resistor	-	30	50	70	$k\Omega$
C_{IO}	Pin capacitance	-	-	5	-	pF

- Guaranteed by design, not tested in production.

5.3.14. ADC characteristics

Table 5-23 ADC characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V_{AVCC}	Power supply	-	1.8	-	5.5	V
I_{DD}	Power consumption	@0.75 MSPS	-	300	-	μA
$C_{IN}^{(1)}$	Internal sample and hold capacitors	-	-	5	-	pF

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
f_{ADC}	Convert clock frequency	$V_{\text{REF}+} = V_{\text{CC}} = 1.8 \sim 2.0 \text{ V}$	0.8	3	6 ⁽²⁾	MHz
		$V_{\text{REF}+} = V_{\text{CC}} = 2.0 \sim 5.5 \text{ V}$	0.8	6	12 ⁽²⁾	
		$V_{\text{REF}+} = V_{\text{REFBUF}}$	0.375	-	0.75	
$t_{\text{samp}}^{(1)}$	Sampling time	$f_{\text{ADC}}=8 \text{ MHz}$	0.438	-	29.94	μs
		$V_{\text{CC}} = 1.8 \sim 2.0 \text{ V}$	3.5	-	239.5	$1/f_{\text{ADC}}$
		$f_{\text{ADC}}=12 \text{ MHz}$	0.292	-	19.96	μs
		$V_{\text{CC}} = 2.0 \sim 5.5 \text{ V}$	3.5	-	239.5	$1/f_{\text{ADC}}$
$t_{\text{samp_setup}}^{(1)}$	Sampling setup time of V_{REFINT}	$V_{\text{REF}+} = V_{\text{CC}} = 2.3 \sim 5.5 \text{ V}$	15	-	-	μs
$t_{\text{conv}}^{(1)}$	Total conversion time	-	-	12	-	$1/f_{\text{ADC}}$
$t_{\text{eoc}}^{(1)}$	Conversion end time	-	-	0.5	-	$1/f_{\text{ADC}}$
$\text{DNL}^{(2)}$	Differential linearity error	-	-	± 2	-	LSB
$\text{INL}^{(2)}$	Integral linearity error	-	-	± 3	-	LSB
Offset ⁽²⁾	Offset error	-	-	± 2	-	LSB

1. Guaranteed by design, not tested in production.
2. Data is based on assessment results and is not tested in production.

5.3.15. Comparator characteristics

Table 5-24 Comparator features⁽¹⁾

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V_{IN}	Input voltage range	-	0	-	$V_{\text{CC}}-1.5$	V
t_{START}	Startup time to reach propagation delay specification	-	-	-	5	μs
t_{D}	Propagation delay	Output low to high	-	-	200	ns
		Output high to low	-	-	150	
V_{offset}	Offset error	-	-	± 5	-	mV
V_{hys}	Hysteresis	No hysteresis	-	0	-	mV
I_{DD}	Consumption	-	-	70	-	μA

1. Guaranteed by design, not tested in production.

5.3.16. Temperature sensor characteristics

Table 5-25 Temperature sensor characteristics

Symbol	Parameter	Minimum	Typical	Maximum	Unit
$T_L^{(1)}$	V_{TS} linearity with temperature	-	± 1	± 2	$^{\circ}\text{C}$
Avg_Slope ⁽¹⁾	Average slope	2.3	2.5	2.7	mV/ $^{\circ}\text{C}$
V_{30}	Voltage at 30 $^{\circ}\text{C}$ ($\pm 5^{\circ}\text{C}$)	0.74	0.76	0.78	V
$t_{\text{START}}^{(1)}$	Start-up time entering in continuous mode	-	70	120	μs
$t_{\text{samp_setup}}^{(1)}$	ADC sampling time when reading the temperature	15	-	-	μs

1. Guaranteed by design, not tested in production.
2. Data is based on assessment results and is not tested in production.

5.3.17. Internal reference voltage characteristics

Table 5-26 Internal reference voltage characteristics (V_{REFINT}) ⁽¹⁾

Symbol	Parameter	Minimum	Typical	Maximum	Unit
V_{REFINT}	Internal reference voltage	1.17	1.2	1.23	V
$t_{\text{start_vrefint}}$	Start time of V_{REFINT}	-	10	15	μs
T_{coeff}	Temperature coefficient of V_{REFINT}	-	-	100 ⁽¹⁾	ppm/ $^{\circ}\text{C}$
I_{VCC}	Current consumption from V_{CC}	-	12	20	μA

1. Guaranteed by design, not tested in production.

5.3.18. ADC internal reference voltage characteristics

Table 5-27 Internal reference voltage characteristics (V_{REFBUF}) ⁽¹⁾

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V_{REF25}	Internal 2.5 V reference voltage	$T_A = 25^{\circ}\text{C}, V_{\text{CC}} = 3.3\text{ V}$	2.425	2.5 ⁽²⁾	2.575	V
V_{REF20}	Internal 2.048 V reference voltage	$T_A = 25^{\circ}\text{C}, V_{\text{CC}} = 3.3\text{ V}$	1.988	2.048 ⁽²⁾	2.108	V
V_{REF15}	Internal 1.5 V reference voltage	$T_A = 25^{\circ}\text{C}, V_{\text{CC}} = 3.3\text{ V}$	1.485	1.5 ⁽²⁾	1.515	V
$T_{\text{coeff_VREFBUF}}$	Temperature coefficient of V_{REFBUF}	$T_A = -40 \sim 85^{\circ}\text{C}$	-	-	120 ⁽¹⁾	ppm/ $^{\circ}\text{C}$
$t_{\text{start_VREFBUF}}$	Start time of V_{REFBUF}	-	-	10	15	μs

1. Guaranteed by design, not tested in production.

5.3.19. COMP internal reference voltage characteristics (4bit DAC)

Table 5-28 Internal reference voltage characteristics (V_{REFCMP}) ⁽¹⁾

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
ΔV_{abs}	Absolute variation	-	-	-	± 0.5	LSB
$t_{start_VREFCMP}$	Start time of internal reference voltage	-	-	10	15	μs

1. Guaranteed by design, not tested in production.

5.3.20. Timer characteristics

Table 5-28 Timer characteristics

Symbol	Parameter	Conditions	Minimum	Maximum	Unit
$t_{res(TIM)}$	Timer resolution time	-	1	-	$t_{TIMxCLK}$
		$f_{TIMxCLK} = 24 \text{ MHz}$	41.667	-	ns
f_{EXT}	Timer external clock frequency on CH1 to CH4	-	-	$f_{TIMxCLK}/2$	MHz
		$f_{TIMxCLK} = 24 \text{ MHz}$	-	12	
Re_{TIM}	Timer resolution	$TIM1/14$	-	16	bit
$t_{COUNTER}$	16 bits counter clock period	-	1	65536	$t_{TIMxCLK}$
		$f_{TIMxCLK} = 24 \text{ MHz}$	0.041667	2730	μs

Table 5-29 LPTIM characteristics (clock selection LSI)

Prescaler	PRESC[2:0]	Minimum overflow value	Maximum overflow value	Unit
/1	0	0.0305	1998.848	ms
/2	1	0.0610	3997.696	
/4	2	0.1221	8001.9456	
/8	3	0.2441	15997.3376	
/16	4	0.4883	32001.2288	
/32	5	0.9766	64002.4576	
/64	6	1.9531	127998.3616	
/128	7	3.9063	256003.2768	

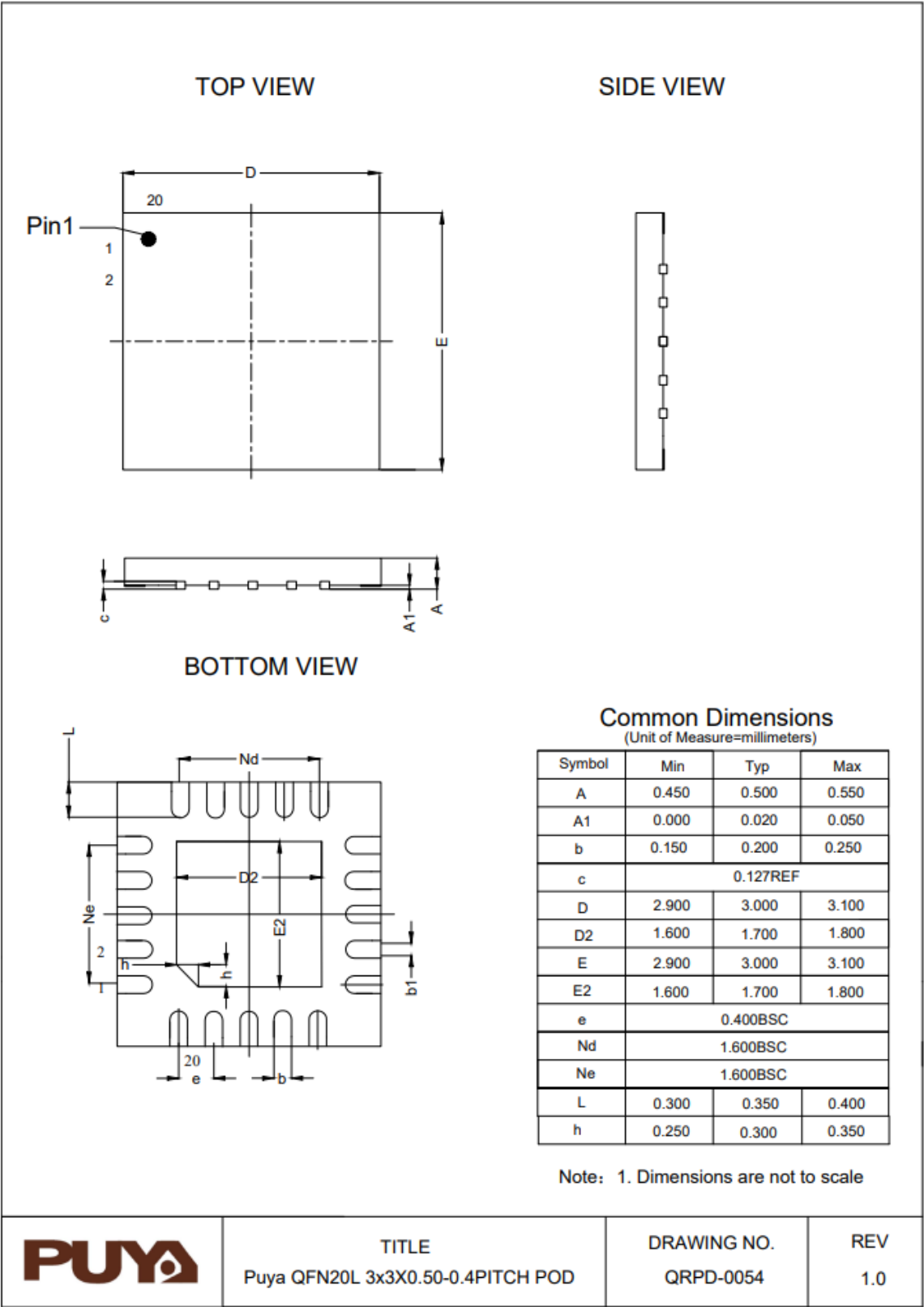
Table 5-30 IWDG characteristics (clock selection LSI)

Prescaler	PR[2:0]	Minimum overflow value	Maximum overflow value	Unit
/4	0	0.122	499.712	ms
/8	1	0.244	999.424	
/16	2	0.488	1998.848	

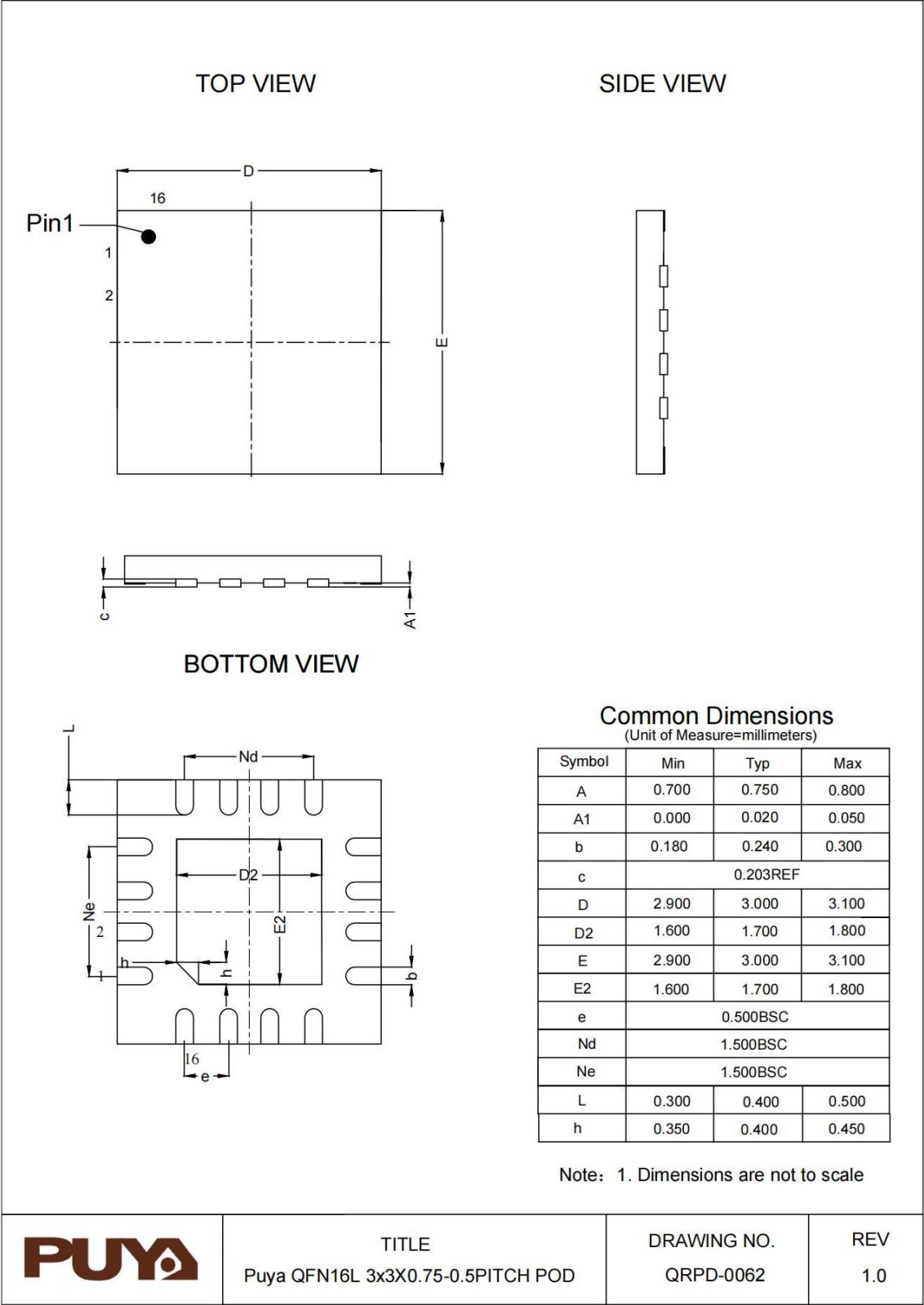
Prescaler	PR[2:0]	Minimum overflow value	Maximum overflow value	Unit
/32	3	0.976	3997.696	
/64	4	1.952	7995.392	
/128	5	3.904	15990.784	
/256	6 or 7	7.808	31981.568	

6. Package information

6.1. QFN20



6.2. QFN16



7. Ordering information

Example:

	PY	32C	64	2	W1	5	U	6	x
Company									
PY = Puya Semiconductor									
Product family									
32bit MCU									
Sub-family									
64 = PY32C64xx									
Product Serial Number									
0-F									
Pin count									
F1 = 20 pins Pinout1									
W1 = 16 pins Pinout1									
User code memory size									
5 = 24 Kbytes									
Package									
U = QFN									
Temperature range									
6 = -40°C to +85°C									
Options									
xxx = code ID of programmed parts(includes packing type)									
TR = tape and reel packing									
TU = Tube Packing									
blank = tray packing									

8. Version history

Version	Date	Description
V0.5	2024.07.19	Initial version



Puya Semiconductor Co., Ltd.

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